Geometric Brownian Motion and Derivative Pricing

A Theoretical Foundation

Aakash Agarwal

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1 Log-Price and the Log-Normal Distribution

1.1 Why Price Is Multiplicative

Asset prices do not evolve additively. A stock moving from \$100 to \$110 and then back to \$100 requires a +10% move followed by a −9.09% move — not a −10% move. The underlying arithmetic is multiplicative: each period's price is the previous period's price scaled by a growth factor.

Over n periods with per-period growth factors $R_1, R_2, \ldots, R_n$:

$$S_n = S_0 \cdot R_1 \cdot R_2 \cdots R_n = S_0 \prod_{i=1}^{n} R_i \tag{1}$$

The product structure creates two problems for direct analysis: (i) the distribution of a product of random variables has no simple closed form, and (ii) prices are bounded below by zero but unbounded above, so a symmetric distribution like the normal cannot be placed directly on S.

1.2 The Log Transformation

Taking the natural logarithm converts multiplicative structure to additive structure:

$$\log S_n = \log S_0 + \sum_{i=1}^{n} \log R_i \tag{2}$$

The log-price is now a *sum* of independent increments. By the Central Limit Theorem, a sum of many independent random variables converges to a normal distribution regardless of the distribution of the individual terms. Therefore:

$$\log S_n \sim \mathcal{N}(\mu_{\log}, \sigma^2_{\log}) \tag{3}$$

which means S_n itself follows a **log-normal distribution**: a random variable whose logarithm is normally distributed.

1.3 Properties of the Log-Normal

If $X = \log S \sim \mathcal{N}(\mu, \sigma^2)$, then:

$$\mathbb{E}[S] = e^{\mu + \frac{1}{2}\sigma^2}, \qquad \mathrm{Var}(S) = \left(e^{\sigma^2} - 1\right) e^{2\mu + \sigma^2} \tag{4}$$

The mean of S is *not* e^{μ} — the $\frac{1}{2}\sigma^2$ correction arises from Jensen's inequality: the exponential function is convex, so $\mathbb{E}[e^X] > e^{\mathbb{E}[X]}$. This correction reappears critically in the GBM solution (Section 4).

The log-normal is bounded below by zero (consistent with limited liability for equity) and has a right-skewed shape with heavier right tails than the normal. Neither property holds for a normal distribution placed directly on S.

2 Linearization via Logarithm

2.1 The Compounding Problem

Percentage returns compound. A position held for T periods with continuously compounding rate r grows to $S_0 e^{rT}$, not $S_0(1+rT)$. Working directly with prices means carrying this nonlinear exponential structure through every calculation. The log transformation removes it.

Define $X = \log S$. Then:

$$S = e^X \quad \Longleftrightarrow \quad X = \log S \tag{5}$$

In X-space, returns are additive and linear. A continuously compounded return of r over time dt is simply $dX = r\,dt$, with no exponential. The dynamics of X can be written, analyzed, and solved using standard linear tools. Results in X-space are then scaled back to price-space via the exponential map $S = e^X$.

2.2 Why This Matters for SDE Analysis

An SDE written directly in S has coefficients that are functions of S — nonlinear. The GBM SDE:

$$dS = \mu S\,dt + \sigma S\,dW \tag{6}$$

has drift μS and diffusion σS — both proportional to S itself. This is a nonlinear SDE in S and has no direct solution.

Transforming to $X = \log S$ via Itô's lemma linearizes the coefficients:

$$dX = \left(\mu - \tfrac{1}{2}\sigma^2\right) dt + \sigma\,dW \tag{7}$$

Both coefficients are now *constants* — the SDE is linear and directly integrable. The $-\frac{1}{2}\sigma^2$ term is the Itô correction: it arises from the second-order term in Itô's lemma and accounts for the curvature of the log function. It has no analogue in ordinary calculus and is the central reason stochastic calculus differs from standard calculus.

The reason linearization enables an integral solution is fundamental. A differential $d(\log S)$ is an infinitesimal change — it describes the instantaneous dynamics but gives no information about total displacement over a finite interval $[0, T]$. To recover $\log S_T - \log S_0$, we must integrate. With constant coefficients, this is immediate:

$$\int_0^T d(\log S) = \int_0^T \left(\mu - \frac{\sigma^2}{2}\right) dt + \int_0^T \sigma\,dW \tag{8}$$

The left side telescopes to $\log S_T - \log S_0$. The first integral on the right is a Riemann integral with constant integrand: $(\mu - \frac{\sigma^2}{2})T$. The second is an Itô integral of a constant: σW_T. The result is exact with no approximation:

$$\log S_T - \log S_0 = \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma W_T \tag{9}$$

If the coefficients were functions of X rather than constants — as they are in the original SDE for S — neither integral would close in this way, and no analytical solution would exist.

2.3 Scaling Back

Once the SDE is solved in log-space, we exponentiate to recover price. This step is necessary because prices are what we observe and trade in the real world — the entire analysis was

conducted in log-space for mathematical convenience, but the output must be expressed in the space of actual dollar prices. Applying $S = e^X$:

$$X_T = X_0 + \left(\mu - \tfrac{1}{2}\sigma^2\right) T + \sigma W_T \quad \Longrightarrow \quad S_T = S_0 \exp\left(\left(\mu - \tfrac{1}{2}\sigma^2\right) T + \sigma W_T\right) \tag{10}$$

Since $X_T = \log S_T$ is normally distributed, S_T is a random variable whose *logarithm* is normal. This is precisely the definition of a **log-normal distribution** — the name comes directly from the relationship: the *log* of S_T is *normal*. Working in log-space and scaling back is not a trick; it is the mathematical statement that price is log-normally distributed.

The $-\frac{1}{2}\sigma^2$ drift correction ensures the expected value of S_T under the physical measure is $S_0 e^{\mu T}$ — consistent with continuous compounding at rate μ. Without it, the expectation would be inflated by the convexity of the exponential.

3 Geometric Brownian Motion — Formulation

3.1 The Discrete Hypothesis

Before writing a continuous SDE, we need a model for how prices move. The starting hypothesis is that log-returns are independent and identically distributed over small intervals. Over one period Δt, the log-return is:

$$\log \frac{S_{t+\Delta t}}{S_t} = \mu \Delta t + \sigma \varepsilon \sqrt{\Delta t}, \qquad \varepsilon \sim \mathcal{N}(0,1) \tag{11}$$

The $\mu \Delta t$ term is the expected return (drift); $\sigma \varepsilon \sqrt{\Delta t}$ is the random shock. The $\sqrt{\Delta t}$ scaling is not arbitrary — it is the correct scaling for Brownian motion increments, ensuring variance scales linearly with time: $\mathrm{Var}(\sigma \varepsilon \sqrt{\Delta t}) = \sigma^2 \Delta t$.

This is structurally a Lévy process in discrete time: increments are independent, stationary, and normally distributed. The sum over n periods gives:

$$\log \frac{S_T}{S_0} = \sum_{i=1}^{n} \left(\mu \Delta t + \sigma \varepsilon_i \sqrt{\Delta t}\right) \sim \mathcal{N}\left(\mu T,\, \sigma^2 T\right) \tag{12}$$

confirming that log-price is normally distributed and price is log-normally distributed. These are not two separate results — they are the same statement written in two different spaces. $\log S_T \sim \mathcal{N}$ is the view in log-space; S_T log-normal is the view in price-space. Exponentiating one gives the other. The log transformation established in Sections 1 and 2 is precisely what connects them.

3.2 The Continuous Limit

As $\Delta t \to 0$, the discrete sum converges to a stochastic integral against Brownian motion W_t. A standard Brownian motion satisfies:

- $W_0 = 0$
- Independent increments: $W_t - W_s \perp W_s - W_r$ for $r < s < t$
- Stationary increments: $W_t - W_s \sim \mathcal{N}(0, t-s)$
- Continuous paths almost surely

The continuous-time limit of the log-return hypothesis is:

$$d(\log S) = \mu\, dt + \sigma\, dW_t \tag{13}$$

Exponentiating (and applying Itô's correction from Section 2) recovers the GBM SDE in price-space:

$$\boxed{dS = \mu S\,dt + \sigma S\,dW_t} \tag{14}$$

3.3 Economic Interpretation of Parameters

Parameter	Meaning	Unit
μ	Drift — expected continuously compounded return	yr^{-1}
σ	Volatility — standard deviation of log-returns	$\text{yr}^{-1/2}$
dW_t	Brownian increment $\sim \mathcal{N}(0, dt)$	—

The drift μ and diffusion σ are both proportional to the current price level S. This is the defining feature of GBM: percentage volatility is constant, not absolute volatility. A stock at \$200 has twice the dollar volatility of the same stock at \$100 at the same σ.

4 GBM SDE — Derivation and Pricing

In GBM, the asset price S_t is both the fundamental stochastic process and the traded asset. A derivative V is written directly on S_t — there is no intermediate layer. The pricing chain is:

$$\boxed{\begin{array}{c} S_t \\ \text{GBM process} \end{array}} \xrightarrow{\text{underlies}} \boxed{\begin{array}{c} V(S,t) \\ \text{Derivative} \end{array}}$$

4.1 Solving the SDE via Itô's Lemma

The GBM SDE $dS = \mu S\,dt + \sigma S\,dW$ is nonlinear in S. To solve it, we apply Itô's lemma to $f(S) = \log S$, which maps the problem into the linear space established in Section 2.

Itô's lemma states: for a twice-differentiable function $f(S)$ and an Itô process dS:

$$df = f'(S)\,dS + \tfrac{1}{2} f''(S)\,(dS)^2 \tag{15}$$

The quadratic variation $(dS)^2$ is computed using the Itô multiplication rules $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt$:

$$(dS)^2 = (\mu S\,dt + \sigma S\,dW)^2 = \sigma^2 S^2\,dt \tag{16}$$

With $f = \log S$, $f' = 1/S$, $f'' = -1/S^2$:

$$d(\log S) = \frac{1}{S}\,dS - \frac{1}{2S^2} \cdot \sigma^2 S^2\,dt \tag{17}$$

$$= \frac{1}{S}(\mu S\,dt + \sigma S\,dW) - \frac{\sigma^2}{2}\,dt \tag{18}$$

$$= \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\,dW \tag{19}$$

Both coefficients are constants. This is a linear SDE with immediate solution by direct integration over $[0, T]$:

$$\log S_T - \log S_0 = \left(\mu - \frac{\sigma^2}{2}\right) T + \sigma W_T \tag{20}$$

Exponentiating:

$$\boxed{S_T = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) T + \sigma W_T\right)} \tag{21}$$

Since $W_T \sim \mathcal{N}(0,T)$, log-price $\log S_T \sim \mathcal{N}\Big(\log S_0 + (\mu - \frac{\sigma^2}{2})T,\ \sigma^2 T\Big)$, confirming the log-normal distribution of S_T.

4.2 European Option Pricing via Risk-Neutral Expectation

The GBM SDE solution gives the distribution of S_T under the physical measure $\mathbb{P}$, where drift is μ. Pricing a derivative requires switching to the **risk-neutral measure** $\mathbb{Q}$, under which all assets earn the risk-free rate r. By Girsanov's theorem, this replaces μ with r in the drift:

$$S_T^{\mathbb{Q}} = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T^{\mathbb{Q}}\right) \tag{22}$$

The no-arbitrage price of any derivative with payoff $\Phi(S_T)$ is:

$$V(S,t) = e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}\Big[\Phi(S_T)\,\Big|\,S_t = S\Big] \tag{23}$$

For a European call with payoff $\Phi(S_T) = \max(S_T - K, 0)$, this expectation is evaluated analytically using the log-normal distribution of $S_T^{\mathbb{Q}}$. Define:

$$d_1 = \frac{\log(S/K) + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t} \tag{24}$$

The integral $e^{-rT}\mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)]$ evaluates to:

$$\boxed{V(S,t) = S\,\mathcal{N}(d_1) - Ke^{-r(T-t)}\mathcal{N}(d_2)} \tag{25}$$

where $\mathcal{N}(\cdot)$ is the standard normal CDF. This is the Black-Scholes formula, derived here via the SDE route. The payoff enters through the boundary of the expectation integral: $S_T > K$ in the region where the call pays out, which defines the integration limits that produce d_1 and d_2.

Limitation. This route prices derivatives of the form $\mathbb{E}^{\mathbb{Q}}[f(S_T)]$ cleanly — payoffs depending only on the terminal price. Path-dependent payoffs (Asian options, barriers, lookbacks) require the joint distribution of S across the entire path, which the terminal solution alone cannot provide. The PDE approach handles these naturally via boundary conditions.

Two routes, one answer. This is a general principle that extends beyond GBM. For any derivative written on any stochastic underlying, there are always two routes to the pricing formula: the **expectation route** — compute $e^{-rT}\mathbb{E}^{\mathbb{Q}}[\text{payoff}]$ using the distribution of the underlying — and the **PDE route** — apply Itô's lemma to V, construct a riskless hedge, and impose no-arbitrage as a boundary value problem. Both routes always produce the same answer. Which one is used in practice depends entirely on which is more tractable for the specific combination of SDE and payoff. For GBM with a European payoff, both are clean — and both are derived in this document. For more complex SDEs or payoffs, one route is often dramatically simpler than the other.

5 The Black-Scholes PDE

5.1 Motivation: Why the SDE Route Is Insufficient

The SDE solution prices simple derivatives cleanly but requires computing $\mathbb{E}^{\mathbb{Q}}[f(S_T)]$ for each payoff function f separately. For standard European options this integral is tractable. For complex payoffs — path-dependent, multi-asset, barrier, or American-exercise — it is not.

The PDE approach is more powerful because it encodes the payoff not in an integral but in a **boundary condition**. Changing the derivative being priced means changing the terminal condition on the same PDE — the equation itself does not change.

5.2 V as a Function of S and t

The value of a derivative at time t depends on the current asset price S and time remaining to expiry. We write $V = V(S, t)$: a deterministic function of the stochastic process S and calendar time t.

Since S follows an Itô process, and V is a smooth function of S and t, Itô's lemma applies directly. This is the key structural observation: V inherits stochastic dynamics from S through the functional relationship $V(S, t)$.

5.3 Applying Itô's Lemma to $V(S, t)$

For $V(S, t)$ with S following GBM, Itô's lemma in two variables gives:

$$dV = \frac{\partial V}{\partial t}\,dt + \frac{\partial V}{\partial S}\,dS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS)^2 \tag{26}$$

Substituting $dS = \mu S\,dt + \sigma S\,dW$ and $(dS)^2 = \sigma^2 S^2\,dt$:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}\right)dt + \sigma S\frac{\partial V}{\partial S}\,dW \tag{27}$$

The dW term makes V stochastic — it cannot be priced directly. The hedge portfolio eliminates it.

5.4 The Hedge Portfolio and No-Arbitrage Condition

Construct a portfolio Π that is long one unit of V and short $\Delta = \partial V/\partial S$ units of S:

$$\Pi = V - \Delta \cdot S = V - \frac{\partial V}{\partial S} \cdot S \tag{28}$$

The instantaneous change in portfolio value is:

$$d\Pi = dV - \frac{\partial V}{\partial S}\,dS \tag{29}$$

Substituting equation (27) and $dS = \mu S\,dt + \sigma S\,dW$, the dW terms cancel exactly:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2}\right)dt \tag{30}$$

The portfolio is now riskless — it has no dW term and its increment is fully determined. A riskless portfolio must earn the risk-free rate r to preclude arbitrage:

$$d\Pi = r\Pi\,dt = r\left(V - S\frac{\partial V}{\partial S}\right)dt \tag{31}$$

5.5 The Black-Scholes PDE

Equating the two expressions for $d\Pi$:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV - rS\frac{\partial V}{\partial S} \tag{32}$$

Rearranging:

$$\boxed{\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0} \tag{33}$$

Two observations. First, μ has disappeared entirely — replaced by r. The drift of S under the physical measure is irrelevant to option pricing; only the risk-free rate and volatility appear. This is the risk-neutral result, arrived at here through no-arbitrage rather than measure change. Second, the PDE holds for *any* derivative written on S — the payoff has not entered yet.

5.6 Boundary Conditions and the Terminal Payoff

The PDE alone has infinitely many solutions. The payoff function selects the unique solution by imposing a terminal condition at $t = T$ and boundary conditions on S.

For a European call with strike K:

$$V(S,T) = \max(S - K,\, 0) \qquad \text{(terminal payoff)} \tag{34}$$

$$V(0,t) = 0 \qquad \text{(worthless if } S \to 0) \tag{35}$$

$$V(S,t) \sim S - Ke^{-r(T-t)} \quad \text{as } S \to \infty \qquad \text{(deep in-the-money limit)} \tag{36}$$

For a European put: replace the terminal condition with $\max(K - S, 0)$. For a barrier option: impose $V = 0$ or $V =$ rebate at the barrier level. The PDE is the same; the boundary conditions encode the contract.

5.7 Analytical Solution via Heat Equation

The Black-Scholes PDE maps to the classical heat equation $\partial u/\partial \tau = \partial^2 u/\partial x^2$ via the substitutions $x = \log(S/K)$, $\tau = \frac{1}{2}\sigma^2(T - t)$, and a scaling transformation on V. The heat equation has a known Green's function solution; inverting the substitutions yields the Black-Scholes formula, which is the same result obtained via the SDE route in Section 4.2:

$$\boxed{V(S,t) = S\mathcal{N}(d_1) - Ke^{-r(T-t)}\mathcal{N}(d_2)} \tag{37}$$

$$d_1 = \frac{\log(S/K) + (r + \frac{\sigma^2}{2})(T - t)}{\sigma\sqrt{T - t}}, \qquad d_2 = d_1 - \sigma\sqrt{T - t} \tag{38}$$

5.8 SDE Route vs PDE Route

	SDE route	PDE route
Itô applied to	$\log S$	$V(S,t)$
Purpose	Solve for S_T distribution	Derive constraint V must satisfy
Payoff enters via	Risk-neutral expectation integral	Terminal boundary condition
Solution method	Direct integration	Heat equation transformation
Handles path-dependence	No	Yes, via modified boundary conditions
Handles American exercise	No	Yes, as a free-boundary PDE

6 Other Widely Used SDEs

GBM is the canonical model but a narrow one. It assumes constant drift, constant volatility, continuous paths, and no mean-reversion. Each of the following SDEs relaxes one or more of these assumptions in response to specific empirical failures of GBM.

6.1 Ornstein-Uhlenbeck Process

The Ornstein-Uhlenbeck (OU) process is the continuous-time analogue of a mean-reverting AR(1). It is the foundation of the Case 1 drift term and the canonical model for any variable that is pulled toward a structural equilibrium.

$$dX = \kappa(\theta - X)\,dt + \sigma\,dW \tag{39}$$

Term	Parameter	Meaning
$\kappa(\theta - X)\,dt$	$\kappa > 0$	Speed of mean-reversion
θ	—	Long-run mean (equilibrium level)
$\sigma\,dW$	$\sigma > 0$	Constant diffusion

Analytical tractability. Unlike GBM, the OU process has a closed-form solution because its drift is linear in X and its diffusion is constant:

$$X_t = \theta + (X_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)}\,dW_s \tag{40}$$

The distribution of X_t is Gaussian for all t, with mean $\theta + (X_0 - \theta)e^{-\kappa t}$ and variance $\frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$. At long horizons, $X_t \to \mathcal{N}(\theta, \frac{\sigma^2}{2\kappa})$ — a stationary distribution.

Half-life. The characteristic timescale of reversion is $\tau_{1/2} = \log(2)/\kappa$. At $t = \tau_{1/2}$, the gap $|X_t - \theta|$ has halved in expectation. This is directly calibratable from data and economically interpretable.

Limitation. The OU process allows X to go negative. For log-price this is acceptable (log-price can be negative). For nominal variables such as interest rates or commodity prices, this is a problem addressed by CIR.

Applications. Interest rate modeling (Vasicek model), commodity log-price, FX log-rate, pairs trading spread, any variable with a structural anchor.

6.2 Cox-Ingersoll-Ross Process

The Cox-Ingersoll-Ross (CIR) process modifies OU by making diffusion proportional to $\sqrt{X}$, which prevents X from reaching zero (under a parameter condition) and keeps it strictly positive.

$$dX = \kappa(\theta - X)\,dt + \sigma\sqrt{X}\,dW \tag{41}$$

The mean-reverting drift is identical to OU. The diffusion $\sigma\sqrt{X}$ is the key modification: it vanishes as $X \to 0$, so the process is repelled from zero. The **Feller condition** $2\kappa\theta \geq \sigma^2$ is sufficient to guarantee $X > 0$ almost surely for all time.

Analytical tractability. CIR has a closed-form transition density in terms of the noncentral chi-squared distribution. This makes MLE calibration and option pricing tractable.

Applications. Interest rate modeling (the original application — short rate in the CIR model), stochastic volatility (the variance process in Heston uses exactly CIR), credit default intensity.

6.3 Heston Stochastic Volatility Model

GBM assumes constant volatility σ. Empirically, implied volatility is not constant — it varies across strikes (the volatility smile) and across time. The Heston model makes volatility itself a stochastic process, driven by a second Brownian motion correlated with the price Brownian motion.

$$dS = \mu S\,dt + \sqrt{v}\,S\,dW_1 \tag{42}$$

$$dv = \kappa(\theta - v)\,dt + \xi\sqrt{v}\,dW_2 \tag{43}$$

$$\mathrm{Corr}(dW_1, dW_2) = \rho\,dt \tag{44}$$

Parameter	Meaning
v_0	Initial variance
κ	Mean-reversion speed of variance
θ	Long-run variance
ξ	Volatility of volatility (vol-of-vol)
ρ	Correlation between price and variance shocks

The variance process dv is exactly CIR (Section 6.2). The correlation ρ is empirically negative for equities: when price drops sharply, implied volatility spikes. This is the *leverage effect*, which Heston captures and GBM does not.

Analytical tractability. Heston (1993) derived a semi-analytical pricing formula for European options using characteristic functions and Fourier inversion. The price is not a simple closed form but requires one numerical integral — far cheaper than full Monte Carlo.

Why it matters over GBM. GBM produces a symmetric, log-normal distribution with constant vol — no smile, no skew. Heston produces a skewed distribution with heavier tails and a volatility smile consistent with observed option markets. The five parameters give the model enough flexibility to fit market prices across strikes and maturities simultaneously.

6.4 Jump-Diffusion Models

Continuous-path SDEs cannot replicate the sudden, large moves observed in financial markets: earnings announcements, central bank decisions, geopolitical shocks. Jump-diffusion models add a compound Poisson process on top of continuous diffusion.

The Merton (1976) model is the canonical form:

$$dS = (\mu - \lambda\bar{J})S\,dt + \sigma S\,dW + S\,dJ \tag{45}$$

where $dJ = (e^J - 1)\,dq$ is the jump component, dq is a Poisson process with intensity λ, and $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$ is the log jump size. The drift is adjusted by $-\lambda\bar{J}$ (where $\bar{J} = e^{\mu_J + \sigma_J^2/2} - 1$ is the expected jump) to keep the expected return at μ.

Parameter	Meaning
λ	Average number of jumps per year
μ_J	Mean log jump size
σ_J	Standard deviation of log jump size

Analytical tractability. Merton derived a closed-form pricing formula for European options as an infinite sum: $V = \sum_{n=0}^{\infty} \frac{e^{-\lambda T}(\lambda T)^n}{n!} \cdot V_{\mathrm{BS}}(\sigma_n)$, where $V_{\mathrm{BS}}(\sigma_n)$ is the Black-Scholes price conditional on n jumps occurring, each with adjusted parameters. In practice the sum converges after 10–20 terms.

Applications. The Case 1 GE SDE uses exactly this jump component (with OU drift instead of GBM drift). Any model of an asset subject to discrete event risk — commodities near OPEC decisions, equities around earnings, EM FX around elections — benefits from an explicit jump term.

6.5 Regime-Switching Models

Regime-switching models allow SDE parameters to change discretely as the market transitions between states (e.g. bull/bear, high/low volatility, crisis/calm). The state variable $Z_t \in \{1, 2, \ldots, K\}$ follows a continuous-time Markov chain with transition intensity matrix Q.

In the simplest two-state version, the asset price follows GBM with parameters that depend on the current regime:

$$dS = \mu_{Z_t} S\, dt + \sigma_{Z_t} S\, dW \tag{46}$$

where (μ_1, σ_1) govern the process in regime 1 and (μ_2, σ_2) in regime 2. The transition between regimes is governed by the generator matrix Q:

$$\mathbb{P}(Z_{t+dt} = j \mid Z_t = i) = q_{ij}\, dt \quad (i \neq j) \tag{47}$$

Why this matters. Financial time series are not stationary. Volatility clusters, trends break, and correlations between assets shift during crises. A single fixed-parameter SDE cannot capture this. Regime switching makes the non-stationarity explicit and tractable: parameters are fixed within each regime and the uncertainty is about which regime is active.

Applications. Case 4 of this project introduces a Markov regime-switching model for a commodity with structurally different supply-demand regimes.

6.6 Summary: GBM and Its Extensions

Each model in this section addresses a specific empirical failure of GBM. The table below organizes them by which assumption they relax.

Model	Assumption relaxed	Key feature	Tractability
GBM	—	Baseline log-normal	Full closed form
OU	No mean-reversion	Structural anchor, Gaussian	Full closed form
CIR	Negative values possible	Positivity via $\sqrt{X}$ diffusion	Noncentral χ^2
Heston	Constant volatility	Stochastic vol + ρ correlation	Semi-analytical (FFT)
Merton	Continuous paths	Compound Poisson jumps	Infinite series
Regime-switching	Stationarity	Markov parameter switching	Numerical / EM

Further Reading

This document covers the theoretical foundations of GBM and its direct extensions. It is the first entry in a systematic series — each subsequent document covers one SDE family with the same structure: hypothesis, derivation, analytical tractability, and implementation. The full series, including code for each case, is maintained in the accompanying repository.